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## VOCALIZING IN THE HOUSE-CAT; A PHONETIC AND FUNCTIONAL STUDY

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A desire to secure new materials for a concrete and factual comparison of animal cries and human speech led the writer to observe and record the vocal sounds of the house-cat. The purpose of this article is to present the methods which have proved, after five years and more of trial and development, to be fairly satisfactory for describing and interpreting these sounds.

The necessity for controlling situation and context, as fully as possible, limited the main study to the writer's own animals; although other subjects were used to check both sound and its interpretation; while any cat-cry which came to ear was welcomed as a test of the method of phonetic recording and of distinguishing essential pattern from incidental and individual variation. Since the *S* which afforded the greatest number of long-term observations, a female given to much vocalizing, was already in her tenth year when the study began, she had long before achieved an equilibrium of acquaintance and habit between herself and her environment, and her vocalizing patterns may therefore be regarded as firmly established.

Of the sixteen phonetic patterns to be distinguished for this mature cat, all but three have been observed in other subjects as well. Two of these ('bewilderment' and 'refusal') are of minor importance, while the third ('acknowledgment') is one which could be expected only from a cat very much at ease in the presence of the auditor. The eight patterns used in mating and fighting ([*'mhrn*], [*'<sup>h</sup>mhrn*], [*'mhrn- $\alpha$ :ou*], [*' $\phi$ - $\phi$ ': $\partial$* ], pain, growl, wail, and snarl) were developed from the vocalizing of dozens of other cats, since these vocal patterns are easily identified and frequently heard from street and yard. Purring and spitting are too familiar to have their existence questioned. The standard onomatopoeic answer the world over to the nursery question "And what does the kitty say?" vouches for the remaining three patterns ('demand,' 'begging demand,' and 'com-

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plaint), all attempts to make distinctions among the cat's *meows*.<sup>1</sup> Although no other serious and systematic study of house-cat vocalization is to be found, indirectly substantiating evidence for the universality of the vocal patterns is furnished by scattered English and foreign literary descriptions of feline behavior even when no attempt has been made to represent phonetically the precise forms of vocalization.

The house-cat, unlike man, has enforced upon it no model of traditional language and no standard of correct pronunciation to which it must conform. Variety of phonetic form in the vocalizing of these animals is due to an unstudied modification of a voiced column of air during inhalation or exhalation, by means of tensional changes in the mouth and throat, by speed and energy of the column of air, and by the extent, time, and rate of the opening of the mouth. Thus vocalization basically derives its form from the constitution and the physiology of the cat. Association with man plays a large part in conditioning the vocalizing situation, but it does not affect the phonetic form, except in so far as such conditioning affects the strength and temper of the vocalizing effort. In other words, this conditioning affects utterance indirectly by way of the cat's somatic condition, not directly through a choice among vocal patterns.

The fact that vocalizing is not artificially stylized in cats into conventional patterns or words means that an indefinitely wide variation of sound and of patterning is possible. Out of this graduated mass the writer has chosen certain points for reference and departure in description. For convenience and familiarity a phonetic alphabet based almost entirely upon the sounds used in general American English has been used to represent the components of the cat's voicings.<sup>2</sup>

Here a caution is called for. The fact that the vocal element in a cat's

<sup>1</sup> "The English cat *mews*, the Indian cat *myans*, the Chinese cat says *mio*, the Arabian cat *naoua*, and the Egyptian cat *mau*. To illustrate how difficult it is to interpret the cat's language, her 'mew' is spelled in thirty-one different ways, five examples being *maeow*, *me-ow*, *mieaou*, *mouw*, and *murr-raow*," I. M. Mellen, *The Science and the Mystery of the Cat*, 1940, 104. It is significant that all these forms are designed to represent an initial nasal sound and a vowel series not varying far from [əu].

<sup>2</sup> The following phonetic symbols have been adopted for recording the vocal sounds of house-cats:

cry affects the ear in much the same way as a like bit of our own speech does not mean that the cat produces the sound as we might. The animal uses the inhaled, as well as the exhaled, breath for vocalizing and does not appear to use the tip of the tongue at all, while its vowel variations are due, for the most part, to changes in throat-tension rather than to modification of the position of the jaws and other mouth-parts. Furthermore, an exact and meaningful description of vocalizing must call attention to tonal elements which, in the recording of man's speech, are left to context or to verbal description. In the analysis which follows, much is still left to such verbal descriptions as 'bright,' 'dull,' 'eager,' 'absent-minded,' 'long,' and 'short.' The possibility of capturing in symbols the precise quality of feline sounds will depend upon the substitution of quantitative measurements for somewhat indefinite terms and upon an increase in our knowledge of the manner in which the various sounds and effects are produced.

### THE THREE MAIN CLASSES OF VOCAL SOUND

The vocal sounds in *Felis domesticus* may first be divided into three main classes. With relation to the mouth these are (1) sounds produced while the mouth is kept closed, the murmurs, (2) sounds produced while the mouth is opened and then gradually closed, resulting in a fixed vowel-pattern, basically [ɑ:ou], and (3) sounds produced while the mouth is held tensely open in one position. Expressed in terms of the voiced column of air these become sounds during which the breath passes through the nose, passes through the mouth, or is forced through the mouth. The differences between these three classes are at bottom differences in degree of intensity of effort. More energy is required to say 'What?' than to murmur 'Mhn?' and still more to snap or thunder 'What!' than to make the calm inquiry.

|                             |                                                     |                               |
|-----------------------------|-----------------------------------------------------|-------------------------------|
| [ɑ] as in <i>father</i>     | [œ] as in French <i>œuf</i>                         | [h] as in <i>balt</i>         |
| [æ] as in <i>hat</i>        | [ø] as in French <i>eux</i>                         | [m] as in <i>men</i>          |
| [ɛ] as in <i>let</i>        | [u] as in <i>food</i>                               | [n] as in <i>men</i>          |
| [ə] as in <i>comma</i>      | [y] as in French <i>vu</i> or<br>German <i>über</i> | [ŋ] as in <i>run</i> <i>g</i> |
| [i] as in <i>machine</i>    |                                                     | [r] as in <i>rat</i>          |
| [j] as in <i>yet</i>        | [f] as in <i>fun</i>                                | [t] as in <i>rat</i>          |
| [o] as in <i>note</i>       | [g] as in <i>go</i>                                 | [w] as in <i>we</i>           |
| [:] indicates prolongation, | ['] indicates inhalation                            |                               |
| [::] extra prolongation     | [?] indicates rising inflection                     |                               |
| [~] indicates nasalization  | [z] indicates wavering or discontinuity             |                               |
| ['] indicates stress-accent |                                                     |                               |

The murmur-pattern is produced by voicing the inhaled or exhaled breath and is composed of an initial impulse represented by [m], a final dying away of breath represented by [n], sometimes sinking to [ŋ], and a more variable central portion, the main rush of breath, represented by [h], which may be given a roll-like flip or trill represented by [r]. The whole may be represented by [mhrn]. With a preceding ['], ['mhrn] may be used to indicate with emphasis that the sound is produced during inhalation.<sup>3</sup> The [hr] roll may be repeated several times within one breath-impulse, thus varying the length of the murmur. Its quality varies first of all according to the presence or absence of the trill; [r] being present, then according to the 'brightness' or 'highness' of the [hr], *i.e.* probably according to tensity. The murmur as a whole may be given two intonations, the one fairly level, the other falling.

In the vowel-pattern the initial [ɑ:] is the most important and variable element, for it receives the full force of the breath-impulse, while the [o] and [u] are merely the inevitable result of the gradual closing of the mouth and consequent narrowing of the voicing aperture. The [o] is purely an incidental form, intermediate between [ɑ] and [u], and is used in transcription to show that the passage from [ɑ] to [u] is continuous. The final [u] may be prolonged for some time by not closing the mouth or shutting off the voiced breath completely and at once. The variations in the initial vowel are variations in tension from relaxed, broad [ɑ] to tense [æ], and to tenser and often nasalized [ɛ]. A major modal transformation of the initial vowel is its shift into [ø] under certain conditions, whereupon the rest of the pattern shrinks to [ə], [ø':ə] instead of [ɑ:ou].

The vowel-pattern is almost never heard alone, and a vowel is seldom the initial sound. Usually the vowel-pattern results from opening the mouth while the sound is already in progress. Even when mouth-opening starts from a closed rest-position, an explosive initial [m]-like sound is heard, as if the impulse to voice the breath were

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<sup>3</sup> The problem of inhalation-exhalation is one which the writer is not able in every case to settle to her complete satisfaction. It appears from clear visual evidence that purring is produced during both inhalation and exhalation and that the ['mhrn] of 'greeting' is an inhaled murmur; but it is not certain whether the other murmur-forms are products of inhalation or of exhalation. Inhalation appears the more probable.

previous to, rather than simultaneous with, the impulse to open the mouth, [mɑ:ou]. If the mouth is already open, the result of further opening is an initial [w], [wɑ:ou]. Most frequently the vowel-pattern is preceded by an inhaled murmur. Thus the basic vowel-pattern cry is the result of opening the mouth upon a current of voiced breath, and the form of the pattern depends upon the vigor and speed of the breath and upon the time and speed of the opening and subsequent closing of the mouth. If the murmur-pattern is completed before the mouth is opened, an [n]- or [ŋ]-like sound precedes the vowel, ['mhrn-ɑ:ou]. At the other extreme, the murmur may be represented by no more than a slight intake of breath, ['mh] or ['m], ['mɑ:ou]. Or the murmur may reach the trilled stage, ['mhr-rɑ:ou]. A rapid opening of the mouth transforms the [r] element into a [j]-like glide. A very rapid opening upon a very tense murmur results in an [i] sound, giving the vowel-pattern cry the form by which the cat's voice is traditionally represented, *i.e.* "miau," "meow," ['mi-ɑ:ou].

Vocalization of the third type, the strained-intensity-patterns, characterized by the mouth held tensely open in one position while the breath is forced through, is too specialized and individual to be represented by a general-model phonetic pattern.

We shall now describe the varieties or patterns of each of these three vocal classes as they appear in the mature cat.

### Murmur Patterns

1. *The Purr*, ['hrn-rhn-'hrn-rhn . . .]. Purring is produced by continuous vibration during both inhalation and exhalation, and it takes its rhythm from the rhythm of breathing. Each stroke is composed of a number of vibrations almost so independently distinguishable as to be counted. The inhaled stroke, represented by ['hrn], is slightly the stronger, *i.e.* the louder and rougher. The smoother exhaled stroke, represented by [rhn], gives continuity. During purred inhalation the throat can be felt vibrating strongly with the pressure of the individual beats perceptible, but during exhalation some more interior part of the throat seems to vibrate, for the vibrations cannot be felt so distinctly and the sound seems more distant and therefore quieter and smoother.

Purring varies in intensity and in duration according to the degree

of interest the occasion has aroused in the cat. On the structural side, purring varies along a roughness-smoothness scale, being roughest when the vibrations are most separately distinguishable. Roughness usually appears at the beginning of a series of purrs, when the purring is most vigorous, and when continued it may be considered a sign of intensity. Smoothness is of two varieties, the one a relaxation and the other an intensification of interest. As the initial impulse to purr fades, the purring becomes quieter, the strokes shorter, more nearly just a slight voicing of ordinary breathing. It is the [r] element, the trill, which drops out. Finally the purring will cease because the cat has fallen asleep or because its attention has been directed toward something else. The other kind of smoothness arises in very eager instead of drowsy purring. A brightness or highness of the [r] element due to an eager or alert attitude brings about an amalgamation of the [r] with the [h] element which results in an almost vowel-like tone making the individual vibrations imperceptible. This is perhaps due to the intensity and prolongation of the individual [r] element as against a repetition of vibratory beats. If, in the course of routine and fairly uninterested purring, the cat should stretch its body or extend its head to sniff at something so that it is thus forced to breathe more deeply, the [hr] will quite accidentally take on this same high, smooth, amalgamated form, which seems to indicate that an attitude of eager interest results in deeper breathing, which in turn affects the form of the purr.

2. *The Level ['mhrn] of Request or Greeting.* In structure the ['mhrn] resembles an isolated and expanded form of the inhaled stroke of purring. The initial [m] indicates its greater body. The quality of the ['mhrn], like that of the purr, involves the presence and brightness of the [r] element. Unlike the ['hrn] of purring, however, ['mhrn] is not composed of a series of separately audible vibrations (which might have been symbolized by writing the purr as ['hrrrn-rrrh n . . .]), but is prolonged by means of repeated new intakes of breath before release through exhalation, represented by repeating the [hr], ['mhr'hr'hrn]. Murmurs of one, two, or three such rolls are most common, although murmuring may be protracted even through the length of time it takes a cat to run up a flight of stairs, continuing a single ['mhr'hr'hr . . . 'hrn] all the way.

3. *The Call*, [<sup>h</sup>mhrn]. A murmur whose initial sound receives a peculiarly tense, constricted emphasis is used to draw someone or something toward the cat. It may vary in tone from soft coaxing to imperative command. The commanding call is a single full-bodied [<sup>h</sup>mhrn!] while the coaxing call is gentler and much repeated, often in series with level [mhrn]s. The superimposed [ə] was chosen to indicate the nature of the call because an attempt to pronounce the colorless schwa vowel [ə] while at the same time beginning the inhaled murmur seems to approximate the effect of the peculiar characteristic of the call. It is perhaps a matter of starting the murmur from some kind of glottal stop instead of from the ordinary relaxed rest-position.

4. *The [mhrŋ] of Acknowledgement or Confirmation*. A short, single murmur with a rapidly falling intonation which changes the final sound from [n] to [ŋ] and frequently causes the omission of the [r] element, making [mhrŋ] seem the more typical form, is used when the cat receives or sees that it is about to receive something it has been seeking.

### Vowel Patterns

1. *The Demand*, [mhrn-ɑ':ou]. The vowel-pattern of demand is subject to much variation. First of all, there is variation due to the position of the main stress, which affects the proportion between the inhaled murmur and the exhaled vowel parts. If the demand is developing from coaxing [mhrn], the mouth may at first not be opened until almost all of the vocalizing impulse has been spent upon the murmur, resulting in [mhrn-ə]. As the demands become more and more vociferous, increasing stress is placed upon the initial vowel while the murmur-element contracts and intensifies from [mhrn] to [mhr] to [mh] to [mhj] to [mi], as for example [mhr-ɑ':ou] or [mi-ɑ':ou]. A relaxing of stress upon the initial vowel comes as failure imposes hopelessness, [mhrn-ɑ:ou].

A second variable element is the quality of the initial vowel. This ranges from low, relaxed [ɑ] to high, tense [æ], with many shadings in between. The voice of one cat can be distinguished from that of another chiefly by means of the difference in their normal initial vowel-sounds, and until this norm has been determined for a cat,



attempts to ascribe significance to particular shades of its vowels have little meaning. The important thing is not the variety uttered by the cat at a particular time but whether that variety represents an increase or decrease of tension from its normal vowel or from immediately preceding vowels.

An infrequent variant of the demand is what may be termed a whisper, ['mhrn-ē']. In a situation in which the cat is aware that it is inadvisable to make a noise and yet cannot overcome the impulse to voice a demand, the result is a quiet demand whose vowel-series has been dampened by nasalization to [ɛ̃].

2. *The Begging Demand*, ['mbrn-a:ou:']. When in pursuing some end the cat becomes so completely absorbed by the goal that all else is lost sight of, the whole demand-pattern becomes slower, more even, and drawn out. The inhaled element is apt to be a mere ['m] due to lack of sufficient vigor to give it more body, the initial vowel tends toward [ɑ] due to relaxation, and the final [u] is prolonged by the slower closing of the mouth.

3. ['maou:ʔ] of *Bewilderment*. Any vowel-pattern in which the initial vowel is heavily stressed indicates high expectancy, a feeling of confidence and right. If the confidence becomes weakened, if the cat sees that events are not going to take their expected course, a note of bewilderment or uncertainty may creep into its vocalizing in the form of a prolonged and rising [u]. Any vowel-pattern whatever, even the most confidently begun demand, may terminate thus if the cat's expectations alter during the cry. When a bewilderment-pattern is followed from the first, it is slow and even in tempo, high in tone, and initiated by ['m], [m], or [w]. While the final [u] receives no stronger a stress than the initial vowel, its prolongation and its rising tone make it predominant. If the initial vowel is an [ɑ], ['maou:ʔ], the impression given is one of bewildered failure of understanding; if the tenser [æ], ['mæou:ʔ], then one of worry.

4. *Complaint*, ['mbrɿ-ɑ:ou]. In its purest form the complaint may be represented by ['mhrɿ-ɑ:ou], the inhaled portion being a short, tense murmur pausing in a glottal stop before the initial vowel. The stress may vary along the confidence-bewilderment scale. A complaint with heavy stress on the initial vowel is a scolding cry. Equal stress and a prolonged final [u] are indicative of some more vaguely formed com-

plaint. The initial vowel may vary from [ɑ] to [æ] according to the intensity of the complaint. When the pause between the murmur and the vowel-series is too short for an [ŋ] to be completed, it is sometimes difficult to distinguish phonetically between complaint and the more hopeless forms of demand. They differ, however, with the direction of search, outwardly toward a specific object during demand and withdrawn inwardly during complaint. Moreover, when complaint forms alone are used, the cat can very often be satisfied with mere verbal sympathy, which is not true of the demand.

5. *The Milder Forms of the Mating Cry*, ['mbrn-ɑ<sub>1</sub>:ou].

6. *The Anger Wail*, [wɑ:ou:]. Since these last two patterns are closely related to two of the strained intensity-patterns, they can best be treated in connection with them.

#### Strained Intensity Patterns

1. *The Growl*, [grrr . . .] and *the Anger Wail*, [wɑ:ou:]. The growling [grrr . . .] is produced low in the tense throat, the breath expelled slowly and steadily through the slightly open, stiffly held mouth. The [g] represents an initial glottal stop which occurs with each fresh breath. Occasionally growling may begin with an [m]. The [rrr . . .] is entirely vocalic and may be prolonged as long as the breath can be made to last. The [r]-like character of the vowel is the effect of the tense condition of the throat and mouth upon what would otherwise probably be a prolonged [Λ].

When two cats are lined up nose to nose, growling away, ready to fight, each waiting for the other to make the first move, the next step vocally is the wail, [wɑ:ou:], a vowel-pattern produced by gradually opening the mouth wider during growling and closing it again to the growl position. This preliminary wailing is long drawn out, the mouth opened and closed slowly and steadily, and the initial vowel tending to an [ɑ]. Wailing, with modification, is continued throughout the fight. It becomes speeded up, and the initial vowel, which is intensified toward [æ], receives a heavy stress whose force is often just a by-product of the physical exertion of fighting.

Slighter wailing, which may be termed angry protest, also occurs occasionally in connection with the growl in highly annoying situations which do not lead to fighting.

2. *The Snarl*, [ $\text{'æ:o}$ ]. During active fighting snarls are also heard. These are rapid inhalations harshly vocalized and marked by a heavy initial intake of breath and stopped suddenly with a slight [o] sound, [ $\text{'æ:o}$ ].

3. *The Mating Cry*, [ $\text{'ø-ø':ə}$ ]. This is a much modified form of the demand. What remains of the demand is the basic rhythm, composed of voiced inhalation, heavily stressed, initial, exhaled vowel and an insignificant trailing off of vowels as the mouth closes. The essential characteristic of the cry is that the initial, exhaled vowel is transformed from [a] to [ø] by a greater dropping of the lower jaw and probably also by the watering of the mouth, for there is much swallowing and licking of chops. The initial vowel takes on a liquid quality which may vary with buccal tensions from an [a] with an underlying [l] quality, [a<sub>l</sub>], through [a<sub>r</sub>] and [æ] to [ø]. In the tensest and most extreme form the voiced inhalation is produced with the mouth slightly open and therefore it becomes a vowel, another [ø], instead of a murmur. At the same time the force of the breath is so completely expended upon the initial exhaled vowel that only an [ə] is heard following it as the mouth closes. The most intense form of the mating cry may therefore be represented by [ $\text{'ø-ø':ə}$ ] and the weakest by [ $\text{'mhrn-a}_1\text{:ou}$ ]. The range between these two extremes is large and continuous. The inhaled part may vary from [ $\text{'mhrn}$ ] to [ $\text{'mhr}$ ] to [ $\text{'hr}$ ] to [ $\text{'æ}$ ] to [ $\text{'ø}$ ] and the exhaled part from [a<sub>l</sub>:ou] to [a<sub>r</sub>:u] to [æ:u] to [ø:ə] with increasing tensivity, force, and loudness given the initial vowel.

4. *Pain*. The shriek of sudden pain, characterized by great strain at mouth and throat and the force of breath, is a tense vowel which may be variously represented by [æ!], [e!], or [i!]. Pain due to an internal cause affects the initial vowel of the slower forms of the complaint-pattern which indicate general discomfort, distorting the initial vowel into a tense, discontinuous, and somewhat nasalized [æzæ]. This prolonged vowel owes its intensity and its wavering or discontinuous character to the force of the opposition which the voiced breath meets from a throat and mouth tensed with the painful state.

5. *The* [ $\text{'æz'æz'æ}$ ] *of Refusal*. This is a low, raspingly discontinuous

sound sometimes uttered as the cat draws back in refusal from something urged upon it.

6. *Spitting*, [fft!]. Spitting is an involuntary reaction when surprised by an enemy or apparent enemy. As the cat changes position with a startle, the breath is forced rapidly through the slightly opened mouth and comes suddenly to a stop, [fft!].

*Vocalization in the kitten.* The vocal patterns which have been outlined above are those of the fully mature cat. Young cats and cats whose vocal essays meet little encouragement will not display so large, so rich, and so varied a range of sounds. The writer has studied twenty-one kittens in nine litters for varying periods of time from birth to seventy-nine days, when (in its twelfth week) the kitten is still far from possessing the vocal virtuosity of the mature cat. Of the murmur-patterns, purring begins at the second day, and the level ['mhrn] of greeting appears during the third week. By the twelfth week ['<sup>a</sup>mhrn] is beginning to be used vaguely, but neither ['mhrn] of request nor ['mhrŋ] of acknowledgement has yet developed. At birth the kitten has a semi-murmur pattern, a grunt represented by [ŋ], which disappears with maturity. It lingers longest in the form [ŋzŋ-ɑ:ou], which is sometimes even heard from a mature cat in bafflement in the face of a stubborn obstacle to movement. In the young kitten the grunt easily closes into purring or opens into a vowel-pattern. The kitten's vowel-pattern, used from birth, is characterized by shrillness and intensity. The inhaled element is vowelized into [i]. The initial vowel varies with age, with the individual and with situational intensity from [y] through [u], [ɛ], [æ] to [ɑ]. The final vowel varies from [i] to the regular [ou] when the less tense initial [ɑ] stage has been reached by all kittens at five or six weeks of age. The vowel-pattern may be preceded by either [m] or [w] according as it is started when the mouth is closed or open. The [w] form is more common with kittens than with mature cats. The whole kitten vowel-pattern, then, may vary from [mi-y] to [mi-ɑ:ou]. This pattern is susceptible of two intonations, distress when the [i] is emphasized, and begging when the initial vowel is stressed and prolonged. The confident demand, the complaint, and the bewilder-

ment-patterns are not yet a part of the repertory at seventy-nine days. It will be noted that these three patterns arise from anticipation, which depends upon experience or conditioning which the kitten has not yet received. Of the strained intensity-patterns, the warning growl will occur as soon as the kitten can run off with a piece of solid food. Spitting appears quite regularly for the first time in the fifth week. No serious fighting takes place and therefore the anger wail and snarling have no opportunity for showing themselves. The very young kitten when pushed away from the breast by another kitten responds with an angry [my!] which may perhaps be considered merely a normal immature intensification of the adult anger-wail. The refusal-pattern, rare even in the mature cat, is never heard, and the mating-patterns must, of course, await sexual maturity.

Compared with the vocalizing of adult cats kitten sounds are chiefly made up, then, of over-intense vowel-patterns and slowly developing murmur-patterns. Progress toward maturity shows itself vocally in mastery of the various murmur-forms and the consequent replacement or modification of earlier vowel-patterns.

The house-cat's vocalization at large may be schematized in the following array of classes and patterns.

#### Mature Cat

- A. *Murmur Patterns* (mouth closed, voiced breath passes through nose)
  1. Purr, ['hrn-rhn-'hrn-rhn . . .]
  2. Level ['mhrn] of Request or Greeting
  3. Call, ['<sup>o</sup>mhrn]
  4. ['mhrŋ] of Acknowledgment or Confirmation
- B. *Vowel Patterns* (mouth opens and gradually closes, voiced breath passes through mouth)
  1. Demand, ['mhrn-α':ou]
  2. Begging Demand, ['mhrn-α:ou]
  3. Bewilderment, ['maou:ʔ]
  4. Complaint, ['mhrŋ-α:ou]
  5. Milder Forms of the Mating Cry, ['mhrn-α<sub>1</sub>:ou]
  6. Anger Wail, [wα:ou:]

C. *Strained Intensity Patterns* (mouth held tensely open, voiced breath forced through mouth)

1. Growl, [grrr . . .]
2. Snarl, ['æ:o]
3. Mating Cry, ['φ-φ':ə]
4. Pain, [æ!]
5. Refusal, ['æz'æz'æ]
6. Spitting, [fft!]

Kitten

(To twelfth week)

A. *Murmur Patterns*

1. Purr
2. Level ['mhrn] of greeting
3. Grunt,

B. *Vowel Patterns*

1. Demand-Wail, [mi-y]-[mi-ɑ:ou]
2. Distress, [mí-y]-[mí-ɑ:ou]
3. Bafflement, [ŋ=ŋ-ɑ:ou]

C. *Strained Intensity Patterns*

1. Growl
2. Spitting
3. Anger, [my!]

### VOCALIZING OCCASIONS AND SITUATIONS

Using our set of functional phonetic characters as a starting-point for describing the vocal sounds of the house-cat, the point of view may now be shifted from the isolation of significant phonetic characteristics to a descriptive account of the main occasions upon which vocalizing is called forth. While these sounds never reach the indicative and 'conversational' level of man's speech, they by no means imply, in the total situation, a one-sided performance but rather a behavioral interchange. The situational aspect may therefore be best treated in terms of complementary relations which lead on to the feline vocalizing that is our central theme. The situation ordinarily involves certain acts-and-events ('goals'), a hominid, or another cat. The first commonly appear in the situation through some hominid agency; but since the cat's apprehension of this agent is often limited, the act or event may be considered, in effect, as a moving body or contrivance.

## Cat-'Goal' Vocalization

The cat may vocalize in connection with a lure or goal whose attainment lies beyond the abilities of *S.* The most common goals are food and the removal of a barrier to free movement, *e.g.* a closed door. Attainment commonly involves the intervention of the hominid, an intervention frequently made effective in the cat by vocalizing, a device which may derive from the young kitten's cry continued until food and warmth are supplied by the mother.

Eight factors can be distinguished as affecting the occurrence and the form of such vocalizing in the cat.

(1) *Inherent value of the goal.* Differences in the inherent value of various goals are reflected in variations in the intensity, quality, and amount of vocalized effort aroused. The clearest examples of this are found in food-preferences.

(2) *Occasional value of the goal.* The occasional value of the goal is given by the strength of the cat's physical need at the moment and is represented by the intensity and the prolongation of its vocalized effort.

(3) *Mere habit, in the absence of real desire or need.* Vocalized goal-seeking behavior is sometimes merely an automatic reaction to a familiar signal connected with a habitual goal. In such circumstances the sounds elicited may be considered as without significance. But since food is nearly always acceptable and since once on the other side of a barrier some object of at least momentary interest is almost invariably found, it is rather difficult to demonstrate conclusively that such a signal has served as a purely automatic release with no relation to need or desire instead of as an announcement of opportunity to achieve a more or less constantly valid end.

(4) *Extent of awareness of a need for specific hominid intervention.* Before a closed door or inaccessible food a young kitten faces only hindrance to free movement or to satisfaction of hunger. With time and repeated experience, however, the nature of the goal and the necessary steps, principally taken by man and leading to achievement, enter into the cat's performance. The relationship becomes cat-man-goal instead of the direct cat-goal. Goal-seeking vocalization is then addressed to man and develops greater variety because it is

directed toward a satisfying, mobile, vocalizing object instead of toward an inert thing. The mature cat coaxes and demands instead of invariably wailing. It is only when the goal exerts so extremely strong an attraction as to absorb the animal completely that the mature cat relapses into the cat-goal relation and a vocalization-pattern (the begging wail) which is not far removed from the kitten's.

(5) *Degree of expectation of achievement.* Time and repeated experience also build up varying degrees of anticipation with respect to different goals and different agents, and these various degrees of expectation are reflected in vocalization. First of all, anticipation affects the amount of effort put forth and therefore the intensity of the initial vocalization. Mute indication of goal-seeking by body position alone—as by sitting in front of a closed door or at the spot where food is usually served—implies fixed anticipation and is not found in the behavior of the young kitten. Secondly, anticipation appears to develop confidence and even assurance which results in a heavy-stress emphasis on the initial vowel of a vowel-pattern in contrast to the prolongation-emphasis of the kitten's wail. In the third place, if, in spite of anticipation, the cat's effort is unsuccessful, the same stress-emphasis will be carried over into the complaint-pattern, *i.e.* vocalization during thwarted expectation appears to differ with the degree of assurance of the anticipated outcome.

(6) *Speed of achievement.* Delay in achievement of a goal induces increased effort in general and increased intensity from mute to murmur to vowel pattern in particular. The form of such delay-induced vocality may possibly be referred to a more or less analytic apprehension of the intervening agent (man) and its expected activities.

(7) *Current mood, physical or emotional state.* The intensity of goal-seeking vocalities may sometimes be due less to the conditions of the immediate situation than to emotive disturbance left over from some previous incident, to physical disturbance caused by ill health, or even to periods of days of especially low or high energy during which all vocalization is shifted slightly toward the mute and murmur patterns or toward the vowel and strained-intensity patterns.

(8) *Success or failure of the effort.* Typical patterns of vocalization are connected with searching out some end, *i.e.* *terminal search*. Suc-



cess, or an act of compliance in man, such as a hand reaching for the door-knob or a dish of food being placed on the floor, is greeted by ['mhrn], ['mhrŋ], or purring. Failure first brings loss of confidence, the weakening of initial vowel-stress, the lengthening of the final vowel, the raising of the prolonged final vowel in bewilderment, and then, as the cat gives up, complaint-patterns varying in stress according to the degree of the original anticipation.

From the discussion of these eight factors, it is readily seen that a single object serving as something to be obtained or attained may at various times arouse various vocalizations, and that unlike objects may arouse similar patterns. One must not expect to find one sound-pattern used constantly for milk, another for meat, a third for door, and the like, as in man's speech.

The cat-goal situation has been described so far only from the cat-toward-goal point of view. A reversed situation is created when a cat is deliberately presented with something desirable. If the cat 'accepts the goal,' some form of the ordinary cat-to-goal situation results. If it does not, the cat refuses it with a show of indifference, or, if the urging and the distaste for it are both intense, with the vocal refusal-pattern, ['æz'æz'æ].

#### Cat-Man Vocalization

Vocalizations which arise when the cat is primarily directed toward a hominid may be divided into three groups according to the thing offered, *i.e.* petting, friendly greeting or sympathy.

The cat's vocal response to petting (head-stroking and verbal admiration) is the purr. A two-day kitten may purr on petting. The hominid is important in the petting-purring situation only as the most satisfactory source or agent. The purr may be obtained in a short time from any strange cat who will stay to be talked to and touched.

The sight or sound of a familiar and welcome person (familiar and welcome as the source of food, petting, material comfort, protection, etc., though none of these is being sought at the moment) is greeted with some simple form of ['mhrn]. Such a situation leads naturally into petting, and the [mhrn]s sink easily into purring. Kit-

tens who have been talked to and petted every day from birth will begin during their third week to leave off feeding and come forward vocalizing toward a familiar voice although at that age the voice has no connection for the kitten with anything but petting.

When a cat is in a disturbed state due to failure to achieve some end, strenuous physical exercise, ill-health, and so on, the disturbance may release sounds of a complaint-pattern, particularly (perhaps always) in the presence of some person, friend or stranger, who shows a willingness to consider or to sympathize. In fact, nearly any unoccupied cat who is not in the state of maximal comfort, as represented by drowsing upon a full stomach in a warm, soft, safe spot with friendly hominids at hand, is in a sufficiently uncomfortable and disturbed state to find ready release through the noises of complaint.

Situations of petting-purring, exchange of friendly greetings, and sympathy-complaint may all be initiated by either cat or man.

In an experiment in inducing cat-man vocalization, every cat met on the street during one month (Oct. 20-Nov. 19) was spoken to in a quiet, sympathetic tone<sup>4</sup> and had its head stroked when that approach was permitted. Thus the cat was offered a friendly greeting, sympathy, and petting. It was discovered that while every cat would turn its head or flick an ear, at least momentarily, to get a glance or a vocal response the speaker must be within six or eight feet at some moment, there must be an absence of disturbing noises, and the cat must not, at the moment, be intently engaged. Out of the twelve responsive cases which occurred, in only four did the cat not accept petting. There were nine cases of the complaint-pattern, five of which were accompanied by near-purr behavior, and one case of purring. Near-purr behavior is that which usually leads to or accompanies purring,—head and tail held high, eyes closed, and a peculiar rubbing of the body, first one side and then the other, against any handy leg or post in a criss-cross manner whether on one spot or in proceeding forward. As was to be expected since the speaker was unfamiliar to all the cats, there was no instance of an ['mhrn] 'greeting' sound.

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<sup>4</sup> Some such expression as 'What's the kitty doing there, mhn?' was used. The familiar sound-pattern 'kitty' served to direct *E*'s greeting toward the cat; the interrogative tone appearing to be effective for a specific response.

The twelve cases are given in detail to illustrate individual variation within a single pattern.

i. ['mhn-æ:ou], once; from a cat who had run a little distance from the spot where it had been sitting staring back at a dog from across the street:

ii. ['mæzæ:ou] repeatedly, once ['mæzæzæ:ou, æ:ou] from a cat locked inside a screen-closed store at night.

iii. Change from flight and fright to near-purr behavior in a cat running by at night.

iv. ['mh-æ:ou] twice faintly, petting, much near-purr behavior, following; from a cat walking by in daytime.

v. One ['mh-æ:ou] after having been spoken to three times; from a cat sitting on a porch who then returned at once to staring toward the direction in which children, including two from his house, were walking away to school.

vi. ['hr-æ:ou], ['hr-æ:ou, ræ:ou] repeated several times, petting; from a cat who tried to lead the observer from the dark store-doorway in which it had been sitting into the lighted store next door.

vii. One ['mæ:ou], petting; from a cat in the doorway of a store not yet opened for the day, going up to the door and looking back at the observer.

viii. ['mɑ:ou] four times, petting, near-purr behavior; from a cat sitting in front of a row of dark stores who then resumed washing himself.

ix. Petting, near-purr behavior; from a cat sleeping in the sun in front of a wide-open store.

x. ['mæzæ:ou] twice, when a cat coming forward for petting with near-purr behavior was stopped by an intervening fence.

xi. Petting, near-purr behavior, purring, following; from a young cat sitting on a doorstep at night; [ŋzŋ] when it discovered that going down another step took it farther from, instead of nearer to, the observer's hand.

xii. ['mæ:ou], very high-pitched, faint; petting, near-purr behavior; from a cat coming forward to be petted from a comfortable, drowsing position on a porch in the daytime.

These vowel-pattern vocalizations all fall phonetically within one type of the complaint because of the insignificance of the inhaled element and because of the lack of stress upon the initial vowel. In

the absence of further information concerning the vocal habits of these cats, it is impossible to say whether the variations in inhaled murmur and in initial vowel were due to individual habit or to the particular situation. Probably both are involved. The effect of situational differences in a similar setting, namely the degree of possibility combined with the degree of desirability of entrance into a store which was the obvious home of the cat, is clearly seen by comparing cases vi to ix. Cat vi, shut out on a cold night from a warm, brightly lighted store, shows the most intense inhaled forms, ['hr] and [r], and a persistence outmatched only by Cat ii, which was screened-in within clear view of freedom. Cat vii, for whom the in-and-out relationship was less sharply contrasted in hope and comfort, shows a lesser intensity and persistence in vocalizing. The relaxed [a] and calm return to washing of Cat viii indicate his reconciliation, in spite of the cold night, to the complete hopelessness of the row of dark closed stores, while Cat ix, comfortably drowsing in the sun with the store-door wide open, cannot rouse himself to the effort of actually voicing a purr, much less a complaint.

#### Cat-Cat Vocalization

Cat giving voice to cat is regulated almost entirely by the life-cycle. During the mating seasons the female sends forth notice of her condition in the loudest and most intense forms of the mating-cry pattern, while the males announce their arrival in less tense forms. 'Invitations' are issued in mild coaxing forms of the ['<sup>h</sup>mhrn] pattern. The mating season brings out the richest tones in the cat's repertory. A shriek (of pain?) and perhaps a low growl from the female mark the moment of coition. Rival males frequently quarrel, using growl-, wail-, and snarl-patterns. All patterns are very much stereotyped as to form and combination and differ from one cat to another in the quality of individual voices. The mating season is over when the female responds to the male's call with a growl instead of a pleasant ['mhr<sub>1</sub>n].

Kittening brings another period of active vocalization. The increase in vocality is not limited to that directed toward the kittens but affects the ordinary daily vocalization for food, exit and greeting as well, since there is greater and more frequent need for food, door-

opening, petting, and sympathy. The mother's regular procedure upon rejoining her kittens is to lick them and call softly with [<sup>1</sup>mhrn]s. In the second month, when the kittens are able to run around and lead increasingly independent lives, the [<sup>1</sup>mhrn] in more commanding tones may be used to try to keep them from wandering too far away or into forbidden territory, or the call may be used in coaxing tones to induce a kitten to play with its mother. If the kitten stops playing while the mother wishes to continue, she may vocalize in complaint-patterns. When a kitten irritates the mother by chewing and clawing her feet and tail, the irritation may find vent in growl and anger-wail. But the effect of this vocalizing upon the kittens' behavior is very slight. The mother must so lie down as to surround the young kittens, she must drag the older ones back by the scruff of the neck, she must wait until the kitten feels like playing with her, she must walk away from the annoying kitten. The voice of man is much more effective in modifying the behavior of kittens even before the kitten has had an opportunity to associate man and the powerful incentive of food. This is due to man's ability to perceive the kittens as individual 'personalities' and therefore to apply vocal encouragement or discouragement at the crucial moment and to support it with reward or punishment far stronger and more individualized than anything the house-cat can offer. The only vocal exchange between cats which occurs with each cat performing equally comes after nine weeks when the kitten is able to greet the mother with [mhrn] upon meeting her and to reply in the same manner to her murmur of greeting. After the kittens are old enough to be interested in solid food, the sound of their mother 'asking for' food may cause them to approach and vocalize similarly, but in this case the mother's vocalizing is a communication no more direct or purposive to the kittens than any such sound as chewing, the rattle of their food dish, or the scratching of a match, which may have become connected for them with the possibility of food.

This study has not included more than one adult cat living in the same household nor more than single kittens beyond nine weeks. Up to that age kittens take no vocal notice of each other except to growl at the approach of one another when eating solid food. Outside the mating periods, the attitude of the most observed *S* of this study

toward any other cat who ventured into her yard was one of hostility, varying in ferocity from mere growling to attack, according to whether she had kittens, to the size of the intruder, and to the possibility of hominid assistance.

Since a cat is limited in the comprehension of its self-object as specifically a 'cat-object,' vocalization arising from the presence of dogs or other animals seems to fall within the non-sexual hostile-friendly aspects of the 'cat-cat' relationship.

The only occurrences of vocalization which cannot be classified within the cat-goal, cat-man, or cat-cat group are those in which vocalization merely accompanies motion without situational reference. A good part of the vocalizing of very young kittens is of this type. In the older cat it is found in the string of murmurings which sometimes accompanies the mild excitement of running in play or in some slight fright which the cat itself has recognized almost instantly as groundless. But if a human auditor is present, even that murmuring may be directed toward him in passing and so may come under cat-man vocalization.

While this situational description and analysis of the house-cat's vocalizing is based on differentiation among the objects or actions which are sought, obtained, or responded to by sounds, the vocalizings are in no way verbal symbols, 'names' for objects and actions. The sounds alone, apart from any situational information, indicate only: (1) Whether the cat is in a state of satisfaction or dissatisfaction (vocalization relaxed or tense), whether it is nearer the purr or nearer the strained intensity patterns; (2) Whether the cat is in a state of certainty and confidence or of uncertainty and bewilderment (strength and position of the main stress); and (3) Whether the cat is trying to induce someone or something to approach or to withdraw ([<sup>h</sup>mhrn] or [grrr . . .] patterns in use). The particular object or action toward which the cat's attention is directed during vocalization is functionally distinguished by means of the habitual, possible, or natural relations holding between the cat and the addressee. In other words, the voicings of the house-cat, unlike most of *our* vocal utterance, is not an attempt verbally to symbolize objects apprehended, inspected or understood, including the self-objects and the body-

states, as well as the multiform relationships maintaining among them all. Rather it is a functional reaction occurring in certain forms under certain conditions. If we seek to make the vocal patterns 'symbolic,' we must regard them as part of the cat's general somatic reaction and not as an expression of the cat's reflective or abstract view of its world.

#### SUMMARY

The domestic cat starts out at birth with bodily facilities for voicing its breath during inhalation and during exhalation, with varying intensities and with the mouth open or closed, facilities which appear to be functionally related (possibly through the autonomic nervous system) to states of satisfaction and unsatisfaction. The house-cat's dependence upon its hominid companions and educators encourages the use and the extension of its vocalizing powers. The form of vocalization at a particular time is dependent upon the condition of the organism, upon the cat's relation to its surroundings, and upon the turn of events. Studies of feline vocalizing, which make full use of situational context, will therefore reveal much concerning the cat's somatic condition and concerning its relations to other organisms, to things, and to events.

These studies are of especial importance in following the cat's development from birth to maturity, since time is required for the attainment of mature vocal patterns. Because the many sounds are specifically correlated with other behavioral patterns, these sounds, which are linear in time and therefore comparatively easy to record, may be conveniently substituted on occasion for studies of more inclusive activities which are multidimensional and therefore impossible of complete simultaneous record.

While the greatest mechanical and phonetical ingenuity may be profitably exercised in recording and analyzing the vocalizing of the cat, the simple fact, instantly and certainly evident to any auditor, that a given sound is of the murmur-, vowel-, or strained-intensity type, is by far the most important point of analysis and is sufficient in many cases to yield significant information.